Homework 15

For Thursday, November 12, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools.

1. Evaluate $\displaystyle\int_1^2 x \ln x \, dx$ by integration by parts.

2. Find $\displaystyle\int \arcsin x \, dx$ for $-1 < x < 1$. Write down your choice of u and dv.

3. Find the function F on $\mathbb{R}$ with $F'(x) = x^2 \sin x$ and $F(0) = 0$.

4. Find $\displaystyle\int e^{\sqrt{x}} \, dx$ for $x > 0$. Begin by setting $t = \sqrt{x}$.

5. Find $\displaystyle\int e^{2x} \cos x \, dx$ by applying integration by parts twice and solving for the original integral.

6. For integers $n \geq 0$, let $L_n = \int_1^e (\ln x)^n \, dx$, with $L_0 = e - 1$. Use integration by parts to express L_n in terms of L_{n-1} for $n \geq 1$. Use your formula to find L_3.

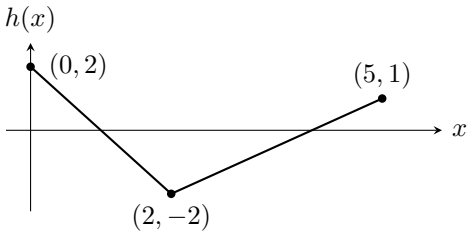
7. Suppose f has a continuous second derivative on $[0, b]$, where $b > 0$. For $0 \leq x \leq b$, prove that

$$f(x) - f(0) - xf'(0) = \int_0^x (x - t) f''(t) \, dt.$$

If $f'' \geq 0$, what does this identity tell you about the graph of f and its tangent at 0?

8. Find the area enclosed by the parabola $y = 4 - x^2$ and the line $y = x + 2$. Determine their intersection points and identify the upper curve between them.

9. For two continuous functions f and g, the graph shows $h(x) = f(x) - g(x)$ on $[0, 5]$. Its three marked points are joined by straight segments. Find both $\int_0^5 h(x)\,dx$ and the area between the graphs of f and g over the same interval.



10. Use horizontal slices to find the area enclosed by $x = y^2$ and $x = 4 - y^2$.

11. Each horizontal slice of a region is a single segment with left endpoint on $x = 0$. The area computed from these slices is

$$\int_0^1 2y\,dy + \int_1^3 (3 - y)\,dy.$$

Sketch the region, label its vertices and write its area as a single integral with respect to x.

12. A horizontal line $y = c$ divides the region $0 \le y \le 2,\ 0 \le x \le y^2$ into two pieces of equal area. Determine c.

13. Suppose $g(x) \le f(x)$ on $[a, b]$, with f, g and h continuous. The curves $y = f(x) + h(x)$ and $y = g(x) + h(x)$ enclose a region together with the lines $x = a$ and $x = b$. Does its area depend on h? Explain using a vertical slice.

14. Let $f : [0, a] \to [0, b]$ be continuous and strictly increasing, with $a, b > 0$, $f(0) = 0$ and $f(a) = b$. The graph $y = f(x)$ divides the rectangle $[0, a] \times [0, b]$ into two regions. Use vertical slices in one region and horizontal slices in the other to prove

$$\int_0^a f(x)\,dx + \int_0^b f^{-1}(y)\,dy = ab.$$

15. A solid has triangular base $0 \le x \le 2,\ 0 \le y \le 2 - x$. Its cross-sections perpendicular to the x-axis are equilateral triangles, each having one side in the base. Find its volume.

16. The base of a solid is $0 \le y \le 2,\ y^2 \le x \le 4$. In the plane at each fixed y, the cross-section is a rectangle. Its bottom edge spans the base, and its height above the base is $1 + y$. Find the volume.

17. The cross-sectional area $S(x)$ of a solid varies linearly between each pair of consecutive measurements:

x (cm)	0	1	3	4
$S(x)$ (cm^2)	0	4	2	2

Find the volume between $x = 0$ and $x = 4$. If S were only known to be continuous, would these four measurements determine the volume? Explain.

18. A square pyramid has its tip at $z = 0$ and base at $z = H$. The square at height z has side az/H, where $a, H > 0$. Divide $[0, H]$ into n equal intervals. Write the volume sum using the larger square in each slab and take its limit. You may use $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$.

19. For $0 \le t \le 1$, the volume of a solid between $x = 0$ and $x = t$ is $V(t) = t + t^2 + t^3/3$. Its cross-sections perpendicular to the x-axis are squares with continuous side length. The xy-plane cuts each square along a *diagonal*. Find the area of the region cut out by the solid in the xy-plane.

20. A solid has square cross-sections perpendicular to the x-axis for $0 \le x \le 1$. The side of each square is a vertical slice of its planar base and has continuous length $w(x)$. The base has area 3. Show that the volume is at least 9, and determine when equality holds. Consider $\int_0^1 (w(x) - 3)^2\, dx$.